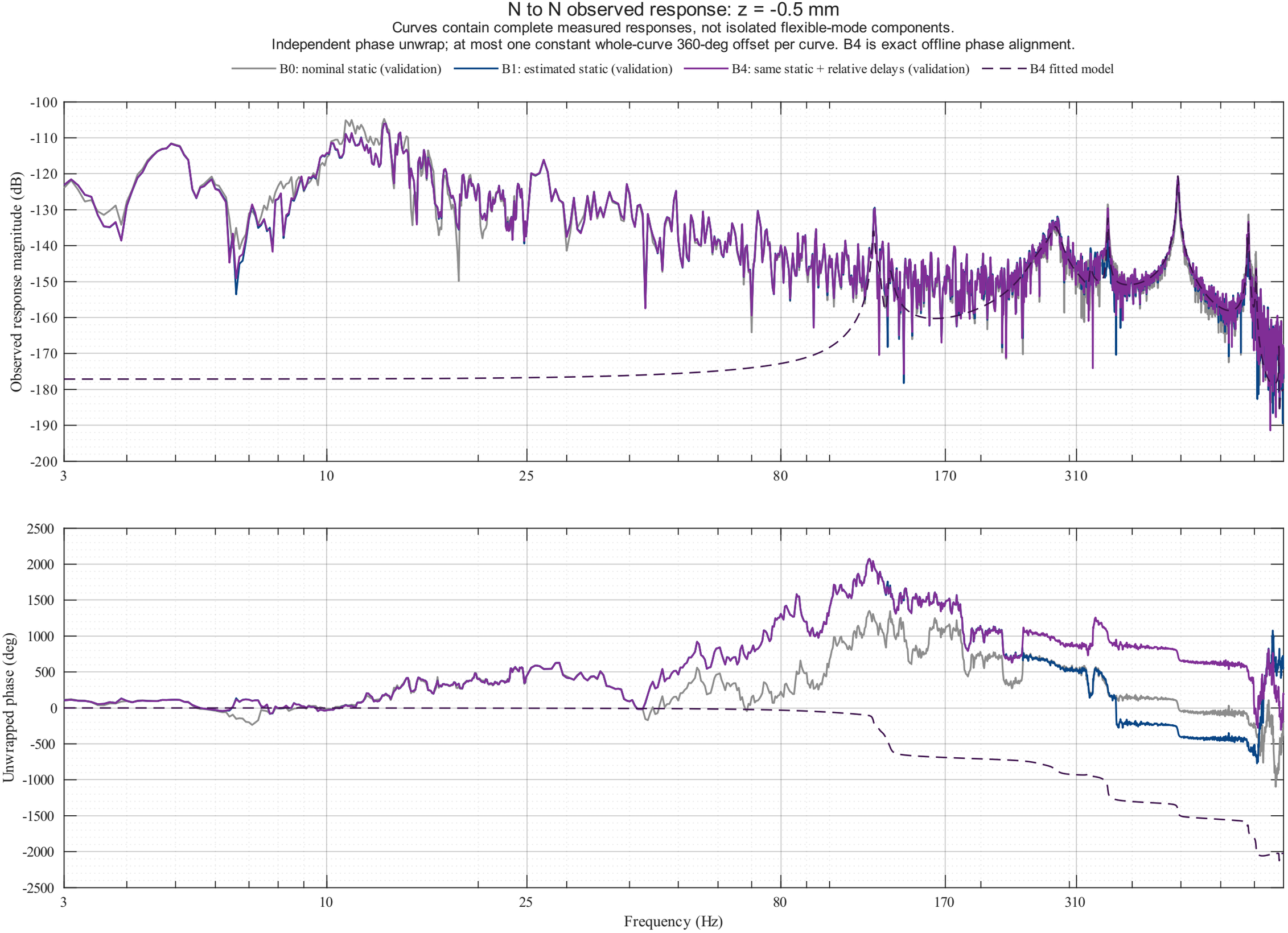

N to N observed response: z = -0.5 mm
Curves contain complete measured responses, not isolated flexible-mode components.
Independent phase unwrap; at most one constant whole-curve 360-deg offset per curve. B4 is exact offline phase alignment.
B0: nominal static (validation)
B1: estimated static (validation)
B4: same static + relative delays (validation)
B4 fitted model
Observed response magnitude (dB)
-100
-110
-120
-130
-140
-150
-160
-170
-180
-190
-200
3
10
25
80
170
310
Unwrapped phase (deg)
2500
2000
1500
1000
500
0
-500
-1000
-1500
-2000
-2500
Frequency (Hz)